

Chapter 6: Statistical Distributions

A Level Statistics Notes · Rexponential

Why this chapter matters

A probability distribution is a model for how likely each possible outcome of a random variable is. This chapter covers the two named distributions the course relies on most heavily: models with a known, standard formula, rather than a bespoke table of individual probabilities. These are the binomial, for counting successes in a fixed number of independent trials, and the Normal, for continuous data clustering symmetrically around a mean. The chapter also covers how to recognise which one (if either) fits a given situation, and both distributions are the direct foundation for the hypothesis tests in Chapter 7.

6.1 Discrete random variables and probability distributions

Concept

A **discrete random variable** takes a countable set of numerical values, each with an associated probability. A **probability distribution** lists every possible value alongside its probability, and since the variable must take exactly one of its possible values, all the probabilities must sum to 1. The **expected value**, $E(X)$, is the long-run average value the variable would take if the experiment were repeated many times — a weighted average of the possible values, each weighted by its own probability.

Key vocabulary

Term	Meaning
Discrete random variable	A variable taking a countable set of numerical values, each with a probability
Probability distribution	A table or function listing every value of X with $P(X = x)$
$E(X)$	The expected value (long-run average) of X

KEY FORMULAS

$$\sum P(X = x) = 1$$
$$E(X) = \sum x P(X = x)$$

Procedure

1. Check that the probabilities given sum to 1 — if one is unknown, this gives an equation to solve for it.
2. To find $E(X)$, multiply each value of x by its probability and sum the results.

3. Interpret $E(X)$ in context as a long-run average, not a value X will necessarily ever take (e.g. an expected value of 3.5 for a die roll is meaningful even though 3.5 is never actually rolled).

Worked example

EXAMPLE 6.1

A discrete random variable X has the probability distribution:

x	1	2	3	4
$P(X = x)$	0.1	k	0.3	0.2

- (a) Find the value of k . (b) Find $E(X)$.

Step 1 (part a). The probabilities must sum to 1. (M1)

$$0.1 + k + 0.3 + 0.2 = 1 \Rightarrow k = 0.4$$

Step 2 (part b). Multiply each x by its probability and sum. (M1)

$$E(X) = 1(0.1) + 2(0.4) + 3(0.3) + 4(0.2) = 2.6$$

Answer (a) $k = 0.4$. (b) $E(X) = 2.6$.

EXAM TIP

It is tempting to treat $E(X)$ as though it must be one of the possible values of X — but it's a weighted average and can (and often does) fall between the actual values the variable can take.

6.2 The binomial distribution

Concept

The **binomial distribution** models the number of "successes" in a fixed number of independent trials, each with the same probability of success. It applies whenever four conditions hold:

- A **fixed number of trials** (n)
- Each trial has only **two outcomes** (success or failure)
- The trials are **independent** of each other
- The probability of success (p) is **constant** across all trials

When all four conditions hold, we write $X \sim B(n, p)$, read " X follows a binomial distribution with parameters n and p ".

Key vocabulary

Term	Meaning
$X \sim B(n, p)$	X follows a binomial distribution with n trials and success probability p

Term	Meaning
The four binomial conditions	The four conditions above: fixed n , two outcomes, independent trials, and constant p

KEY FORMULAS

$$P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}$$

$$E(X) = np, \quad \text{Var}(X) = np(1 - p)$$

Procedure

1. Check the four binomial conditions apply to the scenario (fixed n , two outcomes, independent, constant p) — state this explicitly if asked to justify the model.
2. Identify n and p from the context, and write $X \sim B(n, p)$.
3. For a single value, substitute into the formula, or use a calculator's binomial probability function.
4. For "at least"/"at most"/a range of values, use the cumulative binomial function (or sum individual probabilities), being careful with strict vs non-strict inequalities.

Worked example
EXAMPLE 6.2

A fair coin is tossed 8 times. Let X be the number of heads obtained. (a) State the distribution of X , justifying the conditions. (b) Find $P(X = 5)$. (c) Find $P(X \leq 2)$.

Step 1 (part a). Each toss has two outcomes (head/tail), the probability of a head is constant (0.5) and independent between tosses, and there is a fixed number of trials (8), so $X \sim B(8, 0.5)$. (B1)

Step 2 (part b). Substitute into the binomial formula. (M1)

$$P(X = 5) = \binom{8}{5} (0.5)^5 (0.5)^3 = 0.219 \text{ (3 s. f.)}$$

Step 3 (part c). Use the cumulative binomial distribution (or sum $P(X=0)$, $P(X=1)$, $P(X=2)$). (M1)

$$P(X \leq 2) = 0.145 \text{ (3 s. f.)}$$

Answer (a) $X \sim B(8, 0.5)$. (b) $P(X = 5) \approx 0.219$. (c) $P(X \leq 2) \approx 0.145$.

EXAM TIP

When a question asks to "justify" or "state assumptions for" the binomial model, all four conditions should be mentioned explicitly and linked to the actual context (not just listed generically) — this is commonly a standalone mark, separate from any calculation.

6.3 The Normal distribution

Concept

The **Normal distribution** models continuous data that clusters symmetrically around a mean, with a characteristic bell-shaped curve. It is defined by two parameters: the mean μ and the variance σ^2 , written $X \sim N(\mu, \sigma^2)$. Many naturally occurring continuous variables (heights, weights, measurement errors) are well modelled by a Normal distribution. Because the curve is continuous, $P(X = \text{a specific value})$ is technically 0 — all Normal probabilities are found for a *range* of values. For example, $P(X = 170) = 0$ exactly, but $P(165 < X < 175)$ is a meaningful, non-zero probability.

Key vocabulary

Term	Meaning
$X \sim N(\mu, \sigma^2)$	X follows a Normal distribution with mean μ and variance σ^2
Standard Normal distribution, Z	The Normal distribution with mean 0 and variance 1: $Z \sim N(0, 1)$
Standardising	Converting an X-value to a Z-value to use standard Normal tables

KEY FORMULA

$$Z = \frac{X - \mu}{\sigma}$$

Procedure

1. Identify μ and σ (or σ^2) from the context, and write $X \sim N(\mu, \sigma^2)$.
2. To find a probability, either use a calculator's Normal cumulative distribution function directly with μ and σ , or standardise to Z first using the formula above and use standard Normal tables.
3. Sketch the curve and shade the required region — this avoids sign errors when converting between $P(Z < z)$ and $P(Z > z)$, and helps identify symmetric cases.
4. Use the symmetry of the Normal curve about its mean: $P(Z > z) = P(Z < -z)$ (reflecting the shaded region across the mean), and $P(Z < -z) = 1 - P(Z < z)$ (since the two regions together cover all probability, which sums to 1).

Worked example

EXAMPLE 6.3

The heights of a population of adults are modelled as $X \sim N(170, 8^2)$, in cm. Find the probability that a randomly chosen adult is taller than 180 cm.

Step 1. Standardise. (M1)

$$Z = \frac{180 - 170}{8} = 1.25$$

Step 2. Sketch the curve, shade the area above $z = 1.25$, and read from tables (or calculator). (M1)

$$P(X > 180) = P(Z > 1.25) = 1 - P(Z < 1.25) = 1 - 0.8944 = 0.1056$$

Answer $P(X > 180) \approx 0.106$ (3 s.f.).

EXAM TIP

Always sketch the curve before reading a table value — tables typically give $P(Z < z)$ directly, so any question asking for $P(Z > z)$ or a two-sided range needs a subtraction from 1 (or a symmetry argument) that a sketch makes obvious and a blind table lookup does not.

6.4 Finding probabilities and the inverse Normal

Concept

Section 6.3 found a probability given a value of X . This section reverses that: given a probability (or percentage), find the corresponding value of X . This uses the **inverse Normal** function, and is common in problems involving percentiles, quality control cut-offs, or finding an unknown mean or standard deviation from given probabilities.

Key vocabulary

Term	Meaning
Inverse Normal	Finding the X (or Z) value corresponding to a given cumulative probability
Percentile	The value below which a given percentage of the data falls

Procedure

1. Sketch the curve and shade the described region, to work out whether the required probability is $P(Z < z)$ or needs converting to that form first.
2. Use the inverse Normal function (calculator, or inverse Normal tables) to find the z -value corresponding to that cumulative probability.
3. Convert back to an X -value using $X = \mu + z\sigma$ (rearranging the standardising formula), or if μ or σ is unknown, set up an equation using this rearranged formula and solve.

4. When two unknown parameters are involved (e.g. both μ and σ unknown), use two given probabilities to form two simultaneous equations in μ and σ .

KEY FORMULA

$$X = \mu + z\sigma$$

Worked example
EXAMPLE 6.4

The times taken to complete a puzzle, X minutes, are modelled as $X \sim N(\mu, 5^2)$. It is known that $P(X > 30) = 0.2$. Find the value of μ .

Step 1. Find the z -value such that $P(Z > z) = 0.2$, i.e. $P(Z < z) = 0.8$. (M1)

$$z = 0.8416 \text{ (from inverse Normal)}$$

Step 2. Substitute into $X = \mu + z\sigma$ with $X = 30$, $\sigma = 5$, and solve for μ . (M1)

$$30 = \mu + 0.8416 \times 5 \Rightarrow \mu = 30 - 4.208 = 25.8 \text{ (3 s.f.)}$$

Answer $\mu \approx 25.8$ minutes.

EXAM TIP

Sketch here even more than in 6.3: it is easy to confuse $P(Z > z) = 0.2$ with $P(Z < z) = 0.2$ and read off the wrong side of the inverse Normal table, giving an answer with the wrong sign. A quick sketch makes clear which side of the mean the shaded 20% sits on.

6.5 Choosing a suitable distribution model
Concept

Not every situation fits the binomial or Normal distribution. Part of the assessment is judging which model (if either) is appropriate, and understanding its limitations. The binomial requires discrete counting of successes in fixed, independent, constant-probability trials; the Normal requires a continuous variable that is roughly symmetric about a mean. Recognising when a model's assumptions are violated — and being able to say so — is tested just as often as using the model correctly.

Key vocabulary

Term	Meaning
Model assumption	A condition that must hold for a distribution to be an appropriate model
Model criticism	Identifying why a proposed model may not fit a real situation well

Procedure

1. Identify whether the variable is discrete (countable, e.g. number of successes) or continuous (measured, e.g. length, time, mass).
2. For a discrete counting variable, check the four binomial conditions from 6.2; if any is violated, explain specifically why (e.g. probability of success changes over time, or trials aren't independent).
3. For a continuous variable, consider whether the data is roughly symmetric and bell-shaped, and check whether a Normal model is realistic in context — for example, a Normal model always technically allows negative values, so check whether that makes sense for the variable being modelled.
4. State any conclusion clearly, referring to the specific context rather than giving a generic rule.

Worked example

EXAMPLE 6.5

A researcher wants to model the number of emails a call centre receives per hour using a binomial distribution. Comment on whether this is an appropriate model.

Step 1. Consider the binomial conditions: a binomial distribution needs a *fixed number of trials*, but there is no natural upper limit on the number of emails that could arrive in an hour. (B1)

Step 2. The binomial model is therefore not appropriate; a distribution without a fixed upper bound on the count (such as a Poisson distribution, if covered) would be more suitable. (B1)

Answer The binomial model is not appropriate here, since there is no fixed number of trials — emails can arrive an unlimited number of times in an hour.

EXAM TIP

Model-criticism questions reward specific, context-linked reasoning over generic textbook statements. "The probability might not be constant" earns little on its own; "the probability of a defective item might increase as a machine wears out during the shift" (tied to the actual scenario) earns the mark. Always finish the sentence with why, referring back to the situation in the question.

Chapter summary

Idea	Key point
Discrete random variable	$\sum P(X = x) = 1$; $E(X) = \sum x P(X = x)$
Binomial $X \sim B(n, p)$	Fixed, independent, two-outcome trials with constant p ; $E(X) = np$, $\text{Var}(X) = np(1-p)$
Normal $X \sim N(\mu, \sigma^2)$	Continuous, symmetric; standardise with $Z = (X - \mu)/\sigma$

Idea	Key point
Inverse Normal	Find X or Z from a given probability using $X = \mu + Z\sigma$
Model choice	Check the assumptions of a model against the context before using it

Common exam mistakes

- Listing binomial conditions generically without linking them to the actual scenario when asked to justify a model.
- Reading $P(Z < z)$ from tables when the question actually needs $P(Z > z)$, without sketching first.
- Confusing which side of the mean an inverse Normal probability refers to.
- Treating $E(X)$ as though it must equal one of the possible values of X .

You should now be able to

- Find $E(X)$ for a discrete random variable and use the fact that probabilities sum to 1.
- Identify when a binomial model is appropriate and calculate binomial probabilities.
- Standardise Normal variables, find probabilities, and use the inverse Normal to find unknown values or parameters.
- Critically assess whether a proposed distribution is a suitable model for a given context.